

Setting up your Google IOT

If you'd like to create your own Google IOT account follow this topic.

Installation prerequisites

Install *Portable Git*

Add the 'Portable Git' *mingw32\bin* or *mingw64\bin* (64bit case) directory to your path.

You must have a google account. Go to <https://console.cloud.google.com> and create your Google IOT project, registry, device. Enable APIs, enable API library for PubSub. Create your Pub/Sub topics, subscriptions. Enable MQTT, HTTP for your device and make sure you have a public key in RS256_X509 and in ES256 format. In Google Cloud Console go to API & Services, select Credentials, create Credentials for OAuth Client ID. Download your credentials. You will find here the ClientID and Client secret.

You may follow this documents (without the hardware part)

<https://hub.packtpub.com/build-google-cloud-iot-application/>

<https://codelabs.developers.google.com/codelabs/iotcore-heartrate/index.html?index=..%2F..index#0>

Copy the `gcloud_iot_sample_settings.ini` to the Settings folder (where Settings Folder is in View/Options/Environment Variables) you may rename the file. Modify the values in the ini file.

Using the TINA Google Cloud API

When you create your new project in TINA you have to add the `gcloud_iot.h` file to your project.

You can use the following functions

```
// Mandatory, ini_file_name without the extension
void gcloud_iot_set_param(const char* ini_file_name);

// Publish event (one integer data)
void gcloud_iot_publish_event_i(int a);

// Receive an integer data
bool gcloud_iot_pubsub_pull_i(const char* subscription, int* result);

// Publish event (one integer data)
void gcloud_iot_pubsub_publish_i(const char* topic, int a);
```

`gcloud_iot_set_param` must be called at the begin of the program for example
`gcloud_iot_set_param("gcloud_iot_user_settings");` where `gcloud_iot_user_settings.ini` is your ini file name. You can get the `gcloud_iot.h` file from the *gcloud iot arduino led switch.tsc* circuit file for example.

Compile your program with Atmel Studio or with Arduino.

Using Arduino

Arduino is supported (Arduino MKR1000). On hardware MQTT used, in TINA simulation the HTTP bridge is used. On hardware modify the `ciotc_config.h` to reflect your settings, follow [Google_Cloud_IoT_Core_JWT/README.md](#).

Installing

- Tools/Board/Board manager/
 - Add Arduino SAMD (Cortex-M0+)
- Sketch/Include libraries/Manage libraries, install:
 - Wifi101
 - lwMQTT
 - Google Cloud IoT JWT
- File/Examples/Google Cloud IOT Core JWT

Certificates

Open the FirmwareUpdater in Arduino

- Add domain: mqtt.googleapis.com:443 or mqtt.googleapis.com:8883, click on Upload